

AI Developer / GenAI Engineer

Complete Learning Roadmap · 10–12 Weeks

Your Head Start: With existing experience in MCP server development, RAG systems, ChromaDB, Flask APIs, and healthcare AI — you're already **40–50% of the way there**. Priority gaps: **LangGraph**, multi-agent orchestration, and advanced **GitHub Copilot** features.

Python Foundations

2–3 Weeks

Data Structures & OOP

- Collections, comprehensions, generators
- SOLID principles, abstract base classes, dataclasses
- Factory, Singleton, Observer design patterns
- Pydantic models for data validation

APIs & Async Python

- FastAPI / Flask — REST APIs, middleware, dependency injection
- asyncio, httpx — async/await patterns (critical for agents)
- pytest, mocking, fixtures, coverage reports
- Structured logging with structlog

GenAI & LLM Fundamentals

3–4 Weeks

LLM Concepts

- Prompt engineering: zero-shot, few-shot, chain-of-thought
- RAG — hybrid search, reranking (deepen existing ChromaDB work)
- Fine-tuning basics: LoRA, PEFT — when to fine-tune vs prompt
- Structured outputs, function calling, streaming

Key Libraries

- LangChain — chains, memory, document loaders, vector stores
- LlamaIndex — indexing strategies, query engines, routers
- OpenAI / Anthropic SDK — tool use
- FAISS, Pinecone — vector search exposure

Agent-Based Frameworks ★ HIGH PRIORITY

4–5 Weeks

Agent Fundamentals

- ReAct pattern — Reason + Act loop
- Memory types: short/long-term, episodic
- Planning agents — multi-step reasoning, self-reflection

Frameworks to Master

- LangGraph ★ — stateful multi-agent workflows (top priority)
- CrewAI — role-based multi-agent teams, task delegation
- AutoGen (Microsoft) — conversational agent patterns
- Pydantic AI — structured, type-safe agent development

MCP & A2A Architectures ★ HIGH PRIORITY

2–3 Weeks

Go Deeper (you already have MCP experience)

- MCP — multi-tool servers, resource management, sampling
- Google A2A protocol — inter-agent communication patterns
- Build: multi-agent pipeline where agents call each other via A2A

Practical Projects

- Coding assistant agent with tool use (search, code exec, file I/O)
- Research agent: autonomously search → summarize → draft reports

GitHub Copilot & AI-Assisted Dev ★ HIGH PRIORITY

2 Weeks

Copilot Features • Chat, inline suggestions, workspace agents • Custom instructions via .github/copilot-instructions.md • Building custom Copilot Extensions for domain-specific use	Workflow Integration • GitHub Actions with AI — automated PR reviews, AI-assisted CI/CD • Cursor IDE and Windsurf as supplementary tools • Explore advanced automation and custom rule configurations
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Classical ML, Angular & AWS Fundamentals — Good to Have3–4 Weeks	
Classical ML Basics • Supervised learning — classification, regression, evaluation metrics • Precision, recall, F1, AUC — when each matters • Scikit-learn pipelines, preprocessing, cross-validation • When to use classical ML vs. GenAI approaches DevOps & Deployment • Git branching strategies, PRs, code review workflows • Docker for packaging AI applications • Monitoring: log LLM calls, track token usage, latency	Angular Fundamentals • Components, templates, data binding (one-way & two-way) • Services, dependency injection, RxJS observables • Routing, lazy loading, Angular modules • HttpClient — consuming REST / AI backend APIs • Building AI-powered front-ends (chat UI, dashboards) AWS Fundamentals • Core services: EC2, S3, RDS, Lambda, API Gateway • IAM — roles, policies, least-privilege for AI workloads • SageMaker basics — hosting and invoking ML models • Bedrock — AWS managed GenAI / LLM service • CloudWatch — logging, monitoring, alerting for AI apps • ECS / Elastic Beanstalk — deploying containerised AI services

■ Suggested Weekly Study Plan

Week	Focus Area	Phase	Priority
1–2	Python OOP, async, Pydantic, testing	Phase 1	Foundation
3–4	LLM concepts, LangChain, RAG deepdive	Phase 2	Core GenAI
5–6	LangGraph agents, tool use, memory patterns	Phase 3	★ High Priority
7–8	CrewAI, AutoGen, MCP + A2A integration	Phase 3–4	★ High Priority
9	GitHub Copilot, AI-assisted dev workflows	Phase 5	★ High Priority
10	Classical ML, scikit-learn, model evaluation	Phase 6	Good to have
11	Angular fundamentals — components, services, AI front-end	Phase 6	Good to have
12	AWS core services, SageMaker, Bedrock, IAM, CloudWatch	Phase 6	Good to have
13–14	End-to-end capstone project + Docker + cloud deployment	Phase 6	Capstone

■ Courses	■ Docs to Bookmark	■ Practice
• DeepLearning.AI — LangChain for LLM Apps • DeepLearning.AI — Multi AI Agent Systems • Fast.ai — Practical Deep Learning • Udemy — LangGraph by Eden Marco	• langchain-ai.github.io/langgraph • modelcontextprotocol.io • Google A2A Protocol (GitHub) • Pydantic AI docs	• Build repos on GitHub (recruiters check!) • Kaggle for ML fundamentals • Contribute to open-source agent projects • LeetCode — Easy/Medium Python patterns